

Ayush Kiran Andhale

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A highly motivated Electronics and Telecommunication Engineering student with a minor in Computer Engineering. Strong foundation in programming, embedded systems, and AI-driven IoT solutions. Passionate about applying technical expertise to real-world challenges, with hands-on experience in signal processing, automation, and smart technology development. Seeking opportunities to leverage skills in AI, IoT, and software development to contribute to innovative solutions.

Education

B.Tech. Electronics and Telecommunication Engineering. Bharatiya Vidya Bhavan's Sardar Patel Institute of Technology (SPIT), Mumbai, Maharashtra, India | Year of Completion: 2026

Minor in Computer Engineering : Bharatiya Vidya Bhavan's Sardar Patel Institute of Technology (SPIT), Mumbai, Maharashtra, India | Year of Completion: 2026

Higher Secondary Education (HSC) : Shubhamraje Junior College, Thane, Maharashtra, India | Year of Completion: 2022

Secondary Education (ICSE): Smt. Sulochanadevi Singhanian School, Thane, Maharashtra, India | Year of Completion: 2020

Technical Skills

- Programming Languages: Java, Python, C++
- Databases: PostgreSQL

Projects & Experience

Schneider Electric Infrastructure Limited (SEIL) - *Quality Intern (Ongoing)*

- Specializing in **Process Quality Management**, focusing on optimizing assembly line workflows and ensuring compliance with internal quality standards.
- Gaining hands-on exposure to **Product Testing and Verification** across ring main units and medium-voltage switchgear systems and subassemblies.
- Applying corporate-standard **Quality Control and Root Cause Analysis Tools** (e.g FMEA) to support systematic defect reduction.
- Understanding the development of **Corrective and Preventive Actions (CAPA)** to address recurring quality issues and improve product performance.

Land Cover Classification - *using Machine Learning Algorithms and Data Analytics tools*

- Processed and analyzed Sentinel-2 satellite imagery data for multi-year land cover classification (2017–2025).
- Engineered datasets using band selection, coordinate-based sampling, and class balancing across major land types (urban, water, vegetation, etc.).
- Trained and evaluated multiple machine learning models (SVM, Random Forest, KNN, XGBoost, Naive Bayes) using PCA and standardized features.
- Developed classification maps and statistical comparisons using Python, scikit-learn, Rasterio, and Matplotlib for temporal change analysis.
- Automated model training, prediction, and visualization pipelines integrated with Google Drive and Google Colab for scalable execution.

NEOBIN AI – IoT-Based Smart Waste Segregation, Self-Cleaning & Odour Control System (Ongoing)

- Developed an **AI-driven waste classification system** using MobileNet V2.
- Integrated **IoT-enabled real-time monitoring** with sensors and servo motor automation.
- Designed a **self-sustaining system** using carbon filters for odour control and renewable energy-powered self-cleaning mechanisms.

Extracurricular Activities

- **First Runners-Up – AI in Finance** - Zetheta Algorithms Pvt. Ltd. & Finance & Economics Club, SPIT | March 2025
- **Smart India Hackathon (SIH) 2023** - Selected (Top 7 out of 70 teams) in SPIT's Internal Hackathon for Software Problem Statement.
- **Public Relations Head, Sports Committee, SPIT** - Sep 2023 – Dec 2023 | Led outreach initiatives and event promotions.
- **Subcommittee Member, Sports Committee, SPIT** - Dec 2022 – Sep 2023 | Assisted in organizing and managing sports events.